

Machine Learning (CAI-262)

Objectives:

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- **Types of Regression**
 - **Polynomial Regression**
 - **Ridge Regression**
 - **Lasso Regression**
 - **Logistic Regression**
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Types of Regression

There are several types of regression analysis, including:

1. **Linear Regression:** Linear regression is used to model the relationship between a dependent variable and one or more independent variables, where the relationship is assumed to be linear.
2. **Polynomial regression:** a type of regression in which the independent variable is raised to a power, such as quadratic or cubic regression.
3. **Logistic regression:** used to model the probability of a binary outcome, such as yes or no.
4. **Ridge regression:** a technique used to prevent overfitting in multiple regression models by introducing a penalty for large coefficients.
5. **Lasso regression:** similar to ridge regression but uses a different penalty function that can result in some coefficients being set to zero.
6. **Elastic Net Regression:** Elastic net regression is a combination of ridge and lasso regression and is used when both regularization and variable selection is required.
7. **Time-series regression:** used to analyze time-based data and make predictions about future values.
8. **Bayesian regression:** a type of regression that incorporates Bayesian probability theory to make predictions about the dependent variable.
9. **Nonlinear regression:** involves a dependent variable and one or more independent variables that have a nonlinear relationship.
10. **Poisson regression:** used when the dependent variable is a count or rate. or Poisson regression is used when the dependent variable is a count, and the independent variables are continuous or categorical.

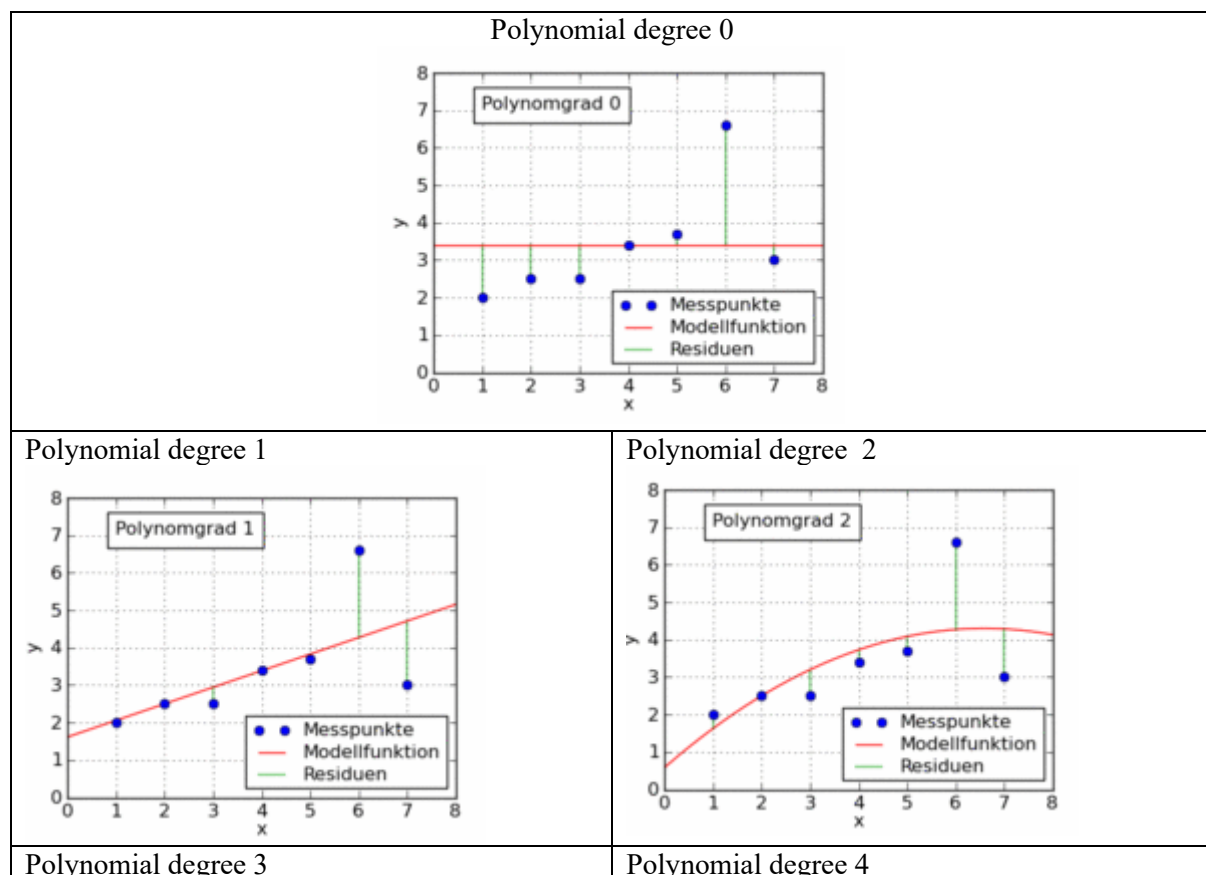
Polynomial regression:

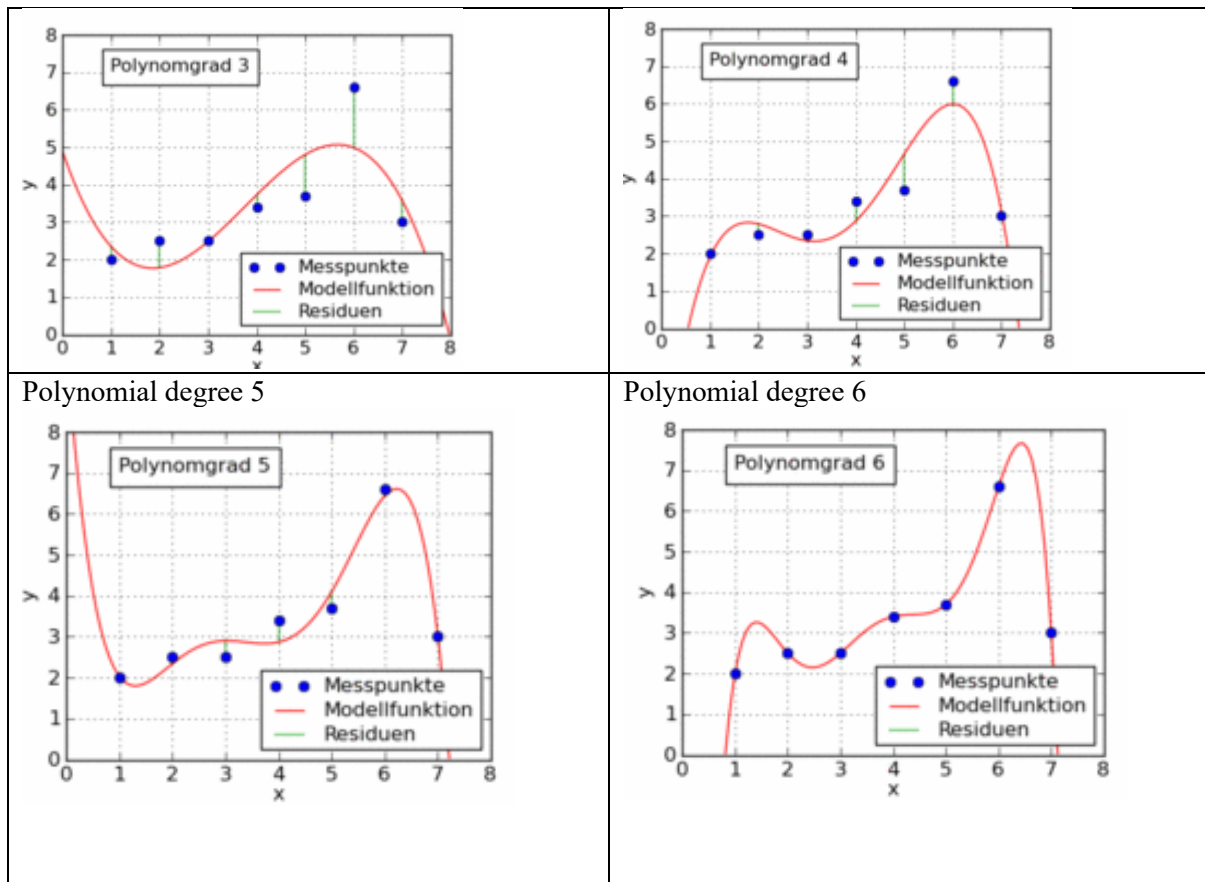
- A polynomial regression model is a machine learning model that can capture non-linear relationships between variables by fitting a non-linear regression line, which may not be possible with simple linear regression.

- It is used when linear regression models may not adequately capture the complexity of the relationship.
- Suppose we have a dataset where variable X represents the Independent data and Y is the dependent data. Before feeding data to a model in the preprocessing stage, we convert the input variables into **polynomial terms** using some degree.
- Consider an example my input value is 35, and the degree of a polynomial is 2, so I will find 35 power 0, 35 power 1, and 35 power 2 this helps to interpret the non linear relationship in data. The equation of polynomials becomes something like this.

$$y = a_0 + a_1x_1 + a_2x_1^2 + \dots + a_nx_1^n$$

- The degree of order which to use is a Hyperparameter, and we need to choose it wisely. But using a high degree of polynomial tries to overfit the data, and for smaller values of degree, the model tries to underfit, so we need to find the optimum value of a degree.





- You can easily understand difference among SLR, MLR and polynomial LR from equations given below.

Simple
Linear
Regression

$$y = b_0 + b_1 x_1$$

Multiple
Linear
Regression

$$y = b_0 + b_1 x_1 + b_2 x_2 + \dots + b_n x_n$$

Polynomial
Linear
Regression

$$y = b_0 + b_1 x_1 + b_2 x_1^2 + \dots + b_n x_1^n$$

Ridge and Lasso Regression

Lasso and ridge regression are two of the most popular variations of linear regression which try to make it a bit more **robust** against outliers, overfitting, and more. Lasso and ridge are very similar, but there are also some key differences between the two that you really have to understand if you want to use them confidently in practice.

The goal of linear regression is to fit a line to a dataset in such a way that it minimizes the mean squared error of our line and our data. With regular linear regression, we don't tell our model anything about how it should achieve this goal. We just want to minimize the MSE and that's it. We don't really care whether the model parameters in our model are very large or very small, we don't care about overfitting, and so on.



$$\text{New Loss 1}(y, y_{pred}) = \text{MSE}(y, y_{pred}) + \alpha \sum_{i=1}^m \theta_i^2 = \text{RidgeMSE}$$

$$\text{New Loss 2}(y, y_{pred}) = \text{MSE}(y, y_{pred}) + \alpha \sum_{i=1}^m |\theta_i| = \text{LassoMSE}$$

The first function is the loss function of ridge regression, while the second one is the loss function of lasso regression. If you are familiar with *norms* in math, then you could say that the lasso penalty is the L1-norm (or Manhattan norm) of our parameter vector θ . This is also the reason why this term is sometimes referred to as an *L1 regularization*. This means that we can rewrite our new loss like this (a norm is denoted using two vertical lines, like this $\|$

$$\text{LassoMSE}(y, y_{pred}) = \text{MSE}(y, y_{pred}) + \alpha \|\theta\|_1$$

Example for difference between lasso and ridge

We will assume that we have some linear model with added regularization. Our linear model has the parameter vector θ with the following values:

$$\theta = \begin{bmatrix} 10 \\ 5 \end{bmatrix}$$

Now let's say that as part of some regularization procedure, we want to reduce the size of our model parameters. For simplicity purposes, let's assume that we'll only change one of the parameters at a time. What happens when we use ridge and lasso regression, respectively? Let's look at both cases.

Ridge

The ridge penalty is $\|\theta\|_2^2 = \theta^T \theta$, so in our case this would be:

$$\text{Ridge Penalty}\left(\begin{bmatrix} 10 \\ 5 \end{bmatrix}\right) = \theta^T \theta = \sum_{i=1}^m \theta_i^2 = 10^2 + 5^2 = 125$$

If we reduce the first parameter by 1, our penalty will now look like this:

$$\text{Ridge Penalty}\left(\begin{bmatrix} 9 \\ 5 \end{bmatrix}\right) = \theta^T \theta = \sum_{i=1}^m \theta_i^2 = 9^2 + 5^2 = 106$$

This means that by reducing our first model parameter from 10 to 9, we reduced our ridge penalty by 19, from 125 to 106. If we instead decreased the other parameter by 1, we would get this penalty:

$$\text{Ridge Penalty}\left(\begin{bmatrix} 10 \\ 4 \end{bmatrix}\right) = \theta^T \theta = \sum_{i=1}^m \theta_i^2 = 10^2 + 4^2 = 116$$

This means that, by reducing our second model parameter from 5 to 4, we reduced our ridge penalty by 9, from 125 to 116. This is *less* than the reduction we got when we reduced our first model parameter!

Since the ridge penalty squares the individual model parameters, large values are taken into account much more heavily than smaller values. This means that our ridge regression model would *prioritize* minimizing large model parameters over small model parameters. This is usually a good thing because if our parameters are already small, they don't need to be reduced even further. Let's now take a look at how this situation looks like when using the lasso penalty.

Lasso

The lasso penalty is $||\theta||_1 = \sum_{i=1}^m |\theta_i|$, so in our case this would be:

$$\text{Lasso Penalty}\left(\begin{bmatrix} 10 \\ 5 \end{bmatrix}\right) = \sum_{i=1}^m |\theta_i| = |10| + |5| = 15$$

If we reduce the first parameter by 1, our penalty will now look like this:

$$\text{Lasso Penalty}\left(\begin{bmatrix} 9 \\ 5 \end{bmatrix}\right) = \sum_{i=1}^m |\theta_i| = |9| + |5| = 14$$

This means that by reducing our first model parameter from 10 to 9, we've reduced our ridge penalty by 1, from 15 to 14. If we instead decreased the other parameter, we would get this penalty:

$$\text{Lasso Penalty}\left(\begin{bmatrix} 10 \\ 4 \end{bmatrix}\right) = \sum_{i=1}^m |\theta_i| = |10| + |4| = 14$$

Aspect	Ridge Regression	Lasso Regression
Regularization Approach	Adds penalty term proportional to square of coefficients	Adds penalty term proportional to absolute value of coefficients
Coefficient Shrinkage	Coefficients shrink towards but never exactly to zero	Some coefficients can be reduced exactly to zero
Effect on Model Complexity	Reduces model complexity and multicollinearity	Results in simpler, more interpretable models
Handling Correlated Inputs	Handles correlated inputs effectively	Can be inconsistent with highly correlated features
Feature Selection Capability	Limited	Performs feature selection by reducing some coefficients to zero
Preferred Usage Scenarios	All features assumed relevant or dataset has multicollinearity	When parsimony is advantageous, especially in high-dimensional datasets
Decision Factors	Nature of data, desired model complexity, multicollinearity	Nature of data, desire for feature selection, potential inconsistency with correlated features

Selection Process	Often determined through cross-validation	Often determined through cross-validation and comparative model performance assessment
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Logistic Regression:

Logistic regression is used for binary classification where we use sigmoid function, that takes input as independent variables and produces a probability value between 0 and 1.

For example, we have two classes Class 0 and Class 1 if the value of the logistic function for an input is greater than 0.5 (threshold value) then it belongs to Class 1 otherwise it belongs to Class 0. It's referred to as regression because it is the extension of linear regression but is mainly used for classification problems.

- Logistic regression predicts the output of a categorical dependent variable. Therefore, the outcome must be a categorical or discrete value.
- It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, it gives the probabilistic values which lie between 0 and 1.
- In Logistic regression, instead of fitting a regression line, we fit an “S” shaped logistic function, which predicts two maximum values (0 or 1).

Logistic regression is the appropriate regression analysis to conduct when the dependent variable is dichotomous (binary). Like all regression analyses, logistic regression is a predictive analysis. It is used to describe data and to explain the relationship between one dependent binary variable and one or more nominal, ordinal, interval or ratio-level independent variables.

Logistic Regression is another statistical analysis method borrowed by Machine Learning. It is used when our dependent variable is dichotomous or binary. It just means a variable that has only 2 outputs, for example, A person will survive this accident or not, The student will pass this exam or not. The outcome can either be yes or no (2 outputs). This regression technique is similar to linear regression and can be used to predict the **Probabilities** for classification problems.

Types of Logistic Regression

Binary logistic regression

Binary logistic regression is used to predict the probability of a binary outcome, such as yes or no, true or false, or 0 or 1. For example, it could be used to predict whether a customer will churn or not, whether a patient has a disease or not, or whether a loan will be repaid or not.

Multinomial logistic regression

Multinomial logistic regression is used to predict the probability of one of three or more possible outcomes, such as the type of product a customer will buy, the rating a customer will give a product, or the political party a person will vote for.

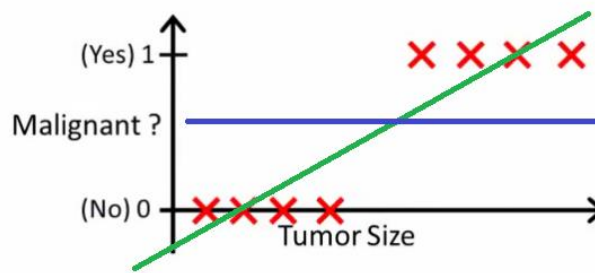
Ordinal logistic regression

is used to predict the probability of an outcome that falls into a predetermined order, such as the level of customer satisfaction, the severity of a disease, or the stage of cancer.

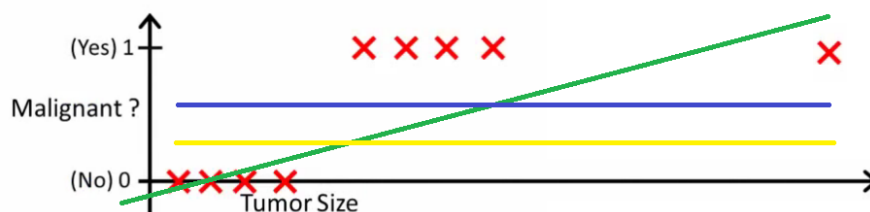
Why do we use Logistic Regression rather than Linear Regression?

After reading the definition of logistic regression we now know that it is only used when our dependent variable is binary and in linear regression this dependent variable is continuous.

The second problem is that if we add an outlier in our dataset, the best fit line in linear regression shifts to fit that point. Now, if we use linear regression to find the best fit line which aims at minimizing the distance between the predicted value and actual value, the line will be like this:



Here the threshold value is 0.5, which means if the value of $h(x)$ is greater than 0.5 then we predict malignant tumor (1) and if it is less than 0.5 then we predict benign tumor (0). Everything seems okay here but now let's change it a bit, we add some outliers in our dataset, now this best fit line will shift to that point. Hence the line will be somewhat like this:



The blue line represents the old threshold and the yellow line represents the new threshold which is maybe 0.2 here. To keep our predictions right we had to lower our threshold value.

Hence we can say that linear regression is prone to outliers. Now here if $h(x)$ is greater than 0.2 then only this regression will give correct outputs.

Another problem with linear regression is that the predicted values may be out of range. We know that probability can be between 0 and 1, but if we use linear regression this probability may exceed 1 or go below 0.

To overcome these problems we use Logistic Regression, which converts this straight best fit line in linear regression to an S-curve using the sigmoid function, which will always give values between 0 and 1. How does this work and what's the math behind this will be covered in a later section?

Assumptions of Logistic regression:

Logistic regression is a statistical method commonly used to analyze data with binary outcomes (yes/no, 1/0) and identify the relationship between those outcomes and independent variables.

Here are some key assumptions for logistic regression:

- **Binary Dependent Variable:** Logistic regression is designed for binary dependent variables. If your outcome has more than two categories, you might need a multinomial logistic regression or other classification techniques.
- **Independent Observations:** The data points should be independent of each other. This means no repeated measurements or clustering within the data.
- **Linearity in the Logit:** The relationship between the independent variables and the logit of the dependent variable ($\ln(p / (1-p))$) is assumed to be linear. This doesn't necessarily mean the outcome itself has a linear relationship with the independent variables, but the log-odds do.
- **No Multicollinearity:** Independent variables shouldn't be highly correlated with each other. Multicollinearity can cause instability in the model and make it difficult to interpret the coefficients.
- **Absence of Outliers:** While not a strict requirement, outliers can significantly influence the model. It's important to check for and address any outliers that might distort the results.
- **Adequate Sample Size:** Logistic regression typically requires a reasonably large sample size to ensure reliable parameter estimates. There are different rules of thumb, but a common guideline is to have at least 10 observations for each independent variable in the model.

Logistic Function

You must be wondering how logistic regression squeezes the output of linear regression between 0 and 1. Let's start by mentioning the formula of logistic function:

$$P = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Best Fit Equation in Linear Regression

We all know the equation of the best fit line in linear regression is:

$$y = \beta_0 + \beta_1 x$$

Let's say instead of y we are taking probabilities (P). But there is an issue here, the value of (P) will exceed 1 or go below 0 and we know that range of Probability is (0-1). To overcome this issue we take "**odds**" of P:

$$P = \beta_0 + \beta_1 x$$

$$\frac{P}{1-P} = \beta_0 + \beta_1 x$$

We know that odds can always be positive which means the range will always be $(0, +\infty)$. Odds are nothing but the ratio of the probability of success and probability of failure. Now the question comes out why did we only take '**odds**'? Because odds are probably the easiest way to do this, that's it. The problem here is that the range is restricted and we don't want a restricted range because if we do so then our correlation will decrease. By restricting the range we are actually decreasing the number of data points and of course, if we decrease our data points, our correlation will decrease. It is difficult to model a variable that has a restricted range. To control this we take the **log of odds** which has a range from $(-\infty, +\infty)$.

$$\log\left(\frac{P}{1-P}\right) = \beta_0 + \beta_1 x$$

Now we just want a function of P because we want to predict probability right? not log of odds. To do so we will multiply by **exponent** on both sides and then solve for P.

$$\exp[\log(\frac{p}{1-p})] = \exp(\beta_0 + \beta_1 x)$$

$$e^{\ln[\frac{p}{1-p}]} = e^{(\beta_0 + \beta_1 x)}$$

$$\frac{p}{1-p} = e^{(\beta_0 + \beta_1 x)}$$

$$p = e^{(\beta_0 + \beta_1 x)} - pe^{(\beta_0 + \beta_1 x)}$$

$$p = p[\frac{e^{(\beta_0 + \beta_1 x)}}{p} - e^{(\beta_0 + \beta_1 x)}]$$

$$1 = \frac{e^{(\beta_0 + \beta_1 x)}}{p} - e^{(\beta_0 + \beta_1 x)}$$

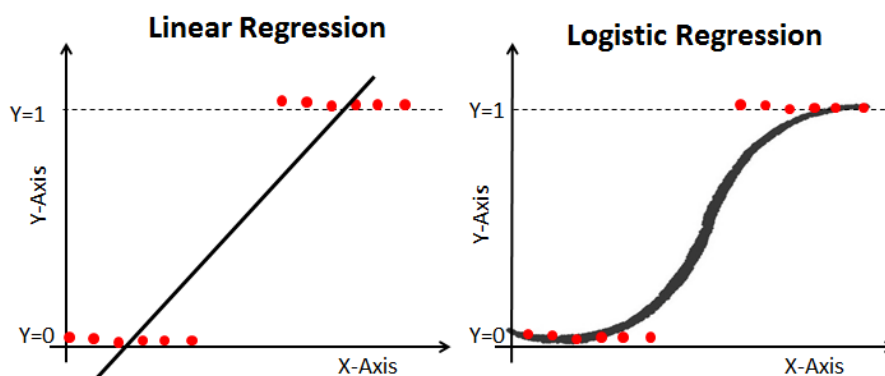
$$p[1 + e^{(\beta_0 + \beta_1 x)}] = e^{(\beta_0 + \beta_1 x)}$$

$$p = \frac{e^{(\beta_0 + \beta_1 x)}}{1 + e^{(\beta_0 + \beta_1 x)}}$$

Now dividing by $e^{(\beta_0 + \beta_1 x)}$, we will get

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}} \text{ This is our sigmoid function.}$$

Now we have our logistic function, also called a **Sigmoid function**. The graph of a sigmoid function is as shown below. It squeezes a straight line into an S-curve.



Differences Between Linear and Logistic Regression

Linear Regression	Logistic Regression
Linear regression is used to predict the continuous dependent variable using a given set of independent variables.	Logistic regression is used to predict the categorical dependent variable using a given set of independent variables.
Linear regression is used for solving regression problem.	It is used for solving classification problems.
In this we predict the value of continuous variables	In this we predict values of categorical variables
In this we find best fit line.	In this we find S-Curve.
Least square estimation method is used for estimation of accuracy.	Maximum likelihood estimation method is used for Estimation of accuracy.
The output must be continuous value, such as price, age, etc.	Output must be categorical value such as 0 or 1, Yes or no, etc.
It required linear relationship between dependent and independent variables.	It not required linear relationship.
There may be collinearity between the independent variables.	There should not be collinearity between independent variables.

Cost Function in Logistic Regression

In linear regression, we use the Mean squared error which was the difference between **y_predicted** and **y_actual** and this is derived from the maximum likelihood estimator. In logistic regression Y_i is a non-linear function ($\hat{Y} = 1 / (1 + e^{-z})$). If we use this in the above MSE equation then it will give a non-convex graph with many local minimas. The problem here is that this cost function will give results with local minima, which is a big problem because then we'll miss out on our global minima and our error will increase.

In order to solve this problem, we derive a different cost function for logistic regression called **log loss** which is also derived from the *maximum likelihood estimation* method.

$$\text{Log loss} = \frac{1}{N} \sum_{i=1}^N -(y_i * \log(\hat{Y}_i) + (1 - y_i) * \log(1 - \hat{Y}_i))$$

In the next section, we'll talk a little bit about the maximum likelihood estimator and what it is used for. We'll also try to see the math behind this log loss function.

In this section, we will try to understand how we can utilize *Gradient Descent* to compute the minimum cost.